



NPTEL Online Certification Course
Generative AI for Computer Vision, IIT Guwahati
Assignment 4: Detailed Solution
Indian Institute of Technology Guwahati



1. What is the main idea behind ResNet?

- a) To reduce image size
- b) To use residual (skip) connections (✓)
- c) To remove convolution layers
- d) To replace activation functions

Answer: b

Detailed Explanation: ResNet introduces residual (skip) connections that help train very deep neural networks effectively.

2. A skip (residual) connection allows information to:

- a) Skip one or more layers (✓)
- b) Skip the input image
- c) Skip the output layer
- d) Skip the loss function

Answer: a

Detailed Explanation: A skip connection directly passes the input of a block to its output, helping preserve information during training.



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3. What is the main advantage of skip connections?

- a) They increase image resolution.
- b) They reduce the number of classes.
- c) They remove pooling layers.
- d) They help reduce the vanishing gradient problem. (✓)

Answer: d

Detailed Explanation: Skip connections improve gradient flow, making it easier to train very deep neural networks.

4. DenseNet differs from ResNet because:

- a) Each layer is connected to every previous layer. (✓)
- b) It has no convolution layers.
- c) It uses only fully connected layers.
- d) It cannot perform image classification.

Answer: a

Detailed Explanation: In DenseNet, every layer receives feature maps from all preceding layers, encouraging feature reuse.



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5. Which CNN architecture is specifically designed for handhold and embedded devices?

- a) AlexNet
- b) VGGNet
- c) MobileNet (✓)
- d) LeNet

Answer: c

Detailed Explanation: MobileNet is designed to be lightweight and computationally efficient for mobile and embedded applications.

6. EfficientNet mainly improves performance by:

- a) Removing convolution layers
- b) Scaling network depth, width, and resolution synchronously (✓)
- c) Using only larger filters
- d) Eliminating activation functions

Answer: b

Detailed Explanation: EfficientNet uses compound scaling to balance network depth, width, and input resolution.



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7. Sequence modeling is mainly used for:

- a) Processing sequential data (✓)
- b) Image compression
- c) Database management
- d) Network security

Answer: a

Detailed Explanation: Sequence modeling handles ordered data such as text, speech, and time-series signals.

8. RNN stands for:

- a) Repeated Neural Network
- b) Residual Neural Network
- c) Recurrent Neural Network (✓)
- d) Random Neural Network

Answer: c

Detailed Explanation: RNN stands for Recurrent Neural Network, which is designed to process sequential data.



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9. Which of the following is a common application of RNN?

- a) Sorting numbers
- b) Image resizing
- c) Data compression
- d) Language translation (✓)

Answer: d

Detailed Explanation: RNNs are widely used for natural language processing, speech recognition, language translation, and time-series prediction.

10. The main feature of an RNN is that it:

- a) Has no hidden layers
- b) Uses only convolution operations
- c) Uses temporal context while processing sequences (✓)
- d) Can process only images

Answer: c

Detailed Explanation: RNNs maintain a hidden state that stores information from previous time steps, enabling them to model sequential dependencies.